Backend task: Budget Tracker API

Build a REST API for a personal budget tracker. One person uses it to record income and expenses, sort them into categories, set a monthly budget per category, and see a summary for each month.

There is no login and only one user. All amounts are in euros. Use any language, framework and database you like. You design the API: endpoints, requests and responses are up to you.

Data

Category

Field	**Rules**
Name	Required. Spaces at the start and end are removed. 1 to 30 characters. Unique, ignoring upper and lower case.
Color	Required. Hex color in the form `#RRGGBB`

Transaction

Field	**Rules**
Type	Required. Income or expense.
Amount	Required. Greater than 0, at most 2 decimals, at most 1,000,000.00
Date	Required. A valid date.
Description	Optional. Up to 200 characters.
Category	Required. Must be an existing category.

Budget

How much the user plans to spend in one category in one month.

Field	Rules
Month	Required
Category	Required. Must be an existing category.
Amount	Same rules as a transaction amount

A category has at most one budget per month.

Features

Categories

- Create a category, change its name or color, and delete it.
- List all categories, sorted by name.
- A category that has transactions cannot be deleted. Deleting a category also deletes its budgets.

Transactions

- Add a transaction, see one, change it and delete it.
- List transactions, newest first. Transactions on the same date: the one added last comes first. Add Pagination
- Filter the list by a date range (both dates included), by category, by type, and by text in the description (ignoring upper and lower case). Filters can be combined.

Import from CSV

The user can upload a CSV file with transactions, for example an export from their bank.

type,amount,date,description,category

expense,12.50,2026-10-08,Lunch,Eating out

income,300.00,2026-10-09,,Other

- The category is given by name and must already exist, ignoring upper and lower case.
- An empty description means there is no description.
- Every row follows the same rules as a transaction.
- A file can have up to 1,000,000 rows.
- After the import, the user knows how many transactions were imported, and which rows failed and why.

Budgets

- Set, change and remove the budget of a category for a month.
- List the budgets of a month.

Monthly summary

For a given month, the user sees:

- total income, total expense, and the balance (income minus expense, can be negative)
- for every category with an expense or a budget in that month: how much was spent, the budget, and how much is left (budget minus spent, negative when over budget). Highest expense first, then by name.

A month with no transactions and no budgets shows zero for every total and no categories.

Errors

Invalid input, things that do not exist, and broken rules (such as a duplicate category name) return an error that explains what is wrong. Bad input never crashes the app.

Also required

- Data is still there after the app restarts.
- Documentation of your API: every endpoint with its input and output, as OpenAPI or in the README.
- Automated tests, at least for the monthly summary, the validation rules and the CSV import.
- A README with how to run the app, the tests and the data command.